Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science andCommerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Practical – III

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Task (on house_price_regression_dataset.csv) (Don't include in print document)
Q.1 Find the correlation between Year_Built and House_Price.

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
# Load dataset
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
# Q1. Correlation between Year_Built and House_Price
correlation = df["Year_Built"].corr(df["House_Price"])
print("Q1. Correlation:", correlation)
```

```
Q1. Correlation: 0.051967361931064535
```

Q.2 Create a correlation matrix for all numerical columns.

```
# Q2. Correlation matrix for all numerical columns
print("\nQ2. Correlation Matrix:")
print(df.corr(numeric_only=True))
```

```
Q2. Correlation Matrix:
                      Square_Footage  Num_Bedrooms  Num_Bathrooms  Year_Built  \
Square_Footage              1.000000     -0.043564      -0.031584   -0.022392
Num_Bedrooms               -0.043564      1.000000       0.022848   -0.015820
Num_Bathrooms              -0.031584      0.022848       1.000000   -0.021063
Year_Built                 -0.022392     -0.015820      -0.021063    1.000000
Lot_Size                    0.089479     -0.009355       0.034923   -0.061050
Garage_Size                 0.030593      0.113761       0.024846   -0.025485
Neighborhood_Quality       -0.008357     -0.049024       0.017585   -0.009549
House_Price                 0.991261      0.014633      -0.001862    0.051967

                      Lot_Size  Garage_Size  Neighborhood_Quality  House_Price
Square_Footage        0.089479     0.030593             -0.008357     0.991261
Num_Bedrooms         -0.009355     0.113761             -0.049024     0.014633
Num_Bathrooms         0.034923     0.024846              0.017585    -0.001862
Year_Built           -0.061050    -0.025485             -0.009549     0.051967
Lot_Size              1.000000     0.002436              0.037630     0.160412
Garage_Size           0.002436     1.000000             -0.011287     0.052133
Neighborhood_Quality  0.037630    -0.011287              1.000000    -0.007770
House_Price           0.160412     0.052133             -0.007770     1.000000
```

Q.3 Perform linear regression using:
X = Year_Built
Y = House_Price
Display the coefficient and intercept.

```
# Q3. Linear Regression: Year_Built vs House_Price
X = df[["Year_Built"]]
Y = df["House_Price"]

model = LinearRegression()
model.fit(X, Y)

print("\nQ3. Linear Regression")
print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)
```

```
Q3. Linear Regression
Coefficient: 638.6524884727304
Intercept: -649854.0823295026
```

Q.4 Calculate the R2 score for the regression model:
Year_Built - House_Price

```
# Q4. R2 Score: Year_Built vs House_Price
Y_pred = model.predict(X)
r2_year = r2_score(Y, Y_pred)

print("\nQ4. R2 Score (Year_Built):", r2_year)
```

```
Q4. R2 Score (Year_Built): 0.0027006067060743044
```

Q.5 Calculate the R2 score for:
Square_Footage - House_Price

```
# Q5. R2 Score: Square_Footage vs House_Price
X = df[["Square_Footage"]]

model_sq = LinearRegression()
model_sq.fit(X, Y)

Y_pred_sq = model_sq.predict(X)
r2_square = r2_score(Y, Y_pred_sq)

print("\nQ5. R2 Score (Square_Footage):", r2_square)
```

```
Q5. R2 Score (Square_Footage): 0.9825992634147099
```

Q.6 Calculate the R2 score for:
Neighborhood_Quality - House_Price

```
# Q6. R2 Score: Neighborhood_Quality vs House_Price
X = df[["Neighborhood_Quality"]]
```

```
model_nq = LinearRegression()
model_nq.fit(X, Y)

Y_pred_nq = model_nq.predict(X)
r2_quality = r2_score(Y, Y_pred_nq)

print("\nQ6. R2 Score (Neighborhood_Quality):", r2_quality)
```

```
Q6. R2 Score (Neighborhood_Quality): 6.037338916142776e-05
```

Q.7 Compare the R2 scores of all three regression models.

```
# Q7. Compare R2 Scores
print("\nQ7. R2 Score Comparison:")
print("Year_Built:", r2_year)
print("Square_Footage:", r2_square)
print("Neighborhood_Quality:", r2_quality)
```

```
Q7. R2 Score Comparison:
Year_Built: 0.0027006067060743044
Square_Footage: 0.9825992634147099
Neighborhood_Quality: 6.037338916142776e-05
```